

ICM-42688-P PIPELINE

Introduction

The ICM-42688-P is a IMU unit which consist of a gyro, accelerometer and a temperature sensor. The device is configurable by I2C, I3C & SPI serial communication, features a 2KB FIFO & 2 programmable interrupts & ultra low power mode. This file will be going over and extracting the needed information to use the ICM-42688-P with a Microcontroller using bare-metal C99.

The device includes industry first 20-bits data format support in FIFO for high data resolution. This FIFO format encapsulates 19-bits of gyroscope data and 18- bits of accelerometer data.

Parameter	ICM-40607 Sensorhub	ICM-42605 Sensorhub	ICM-42686-P Handheld Action	ICM-42688-P HMD & Robotics
GYRO Noise (mdps/rt-Hz)	7	3.8	5.3	2.8
GYRO Offset Temp Stability (mdps/°C)	±30	±20	±10	±5
GYRO Range & Resolution	±2000 dps; 16- bits	±2000 dps; 16- bits	±4000 dps; 16/19- bits	±2000 dps; 16/19- bits
ACCEL Noise (µg/rt-Hz)	110	70	70	AX: 65; AZ: 70
ACCEL Range & Resolution	±16 g; 16-bits	±16 g; 16-bits	±32 g; 16/18-bits	±16 g; 16/18-bits
ODR & Sample Sync	8 kHz; No RTC	8 kHz; No RTC	32 kHz; RTC	32 kHz; RTC

Features

Gyroscope features:

- Digital output (x,y,z) angular rates with programmable scales of ±15.625, ±31.25, ±62.5, ±125, ±250, ±1000, ±2000 deg/sec
- Low Noise (LN) & Low power (LP) modes supported
- Digitally-programmable low-pass filters
- Factory calibrated sensitivity scale factor
- self test

Accelerometer features:

- Digital output (x,y,z) with programmable full scale (±2g, ±4g, ±8g and ±16g)
- Low Noise (LN) & Low power (LP) modes supported
- User-Programmable interrupts
- Wake on motion interrupt

Motion features:

- Pedometer: tracks step count, also issues step detect interrupt
- Tilt detection: issues interrupt when the tilt angle exceeds 35 deg
- Raise to wake/sleep
- Tap detection
- Wake on motion

Additional features:

- External clock input, reduce system level sensitivity error.
- 2k byte fifo buffer enables the applications processor to read in burst
- 20-bits data format support in FIFO for high data resolution
- User programmable digital filters for gyro, Acc and temperature sensor
- 12.5 mHz I3C / 1MHz I2C & 24MHz SPI
- digital output temp sensor
- 20,000 g shock tolerant
- MEMS structure hermetically sealed and bonded at wafer level
- RoHS and Green compliant.

SPI interface

SPI will be the chosen communication protocol between the sensor and the controller for reliability and higher serial data output (SDO). The ICM-42688-P supports 3 & 4 wire SPI for the host interface, the device always operates as slave device during Master-Slave SPI operation.

SPI Operational features:

1. data is delivered MSB first and LSB last (big endian)
2. Data is latched on rising edge of SCLK
3. Data is transitioned on falling edge of SCLK
4. Maximum frequency is 24MHZ
5. SPI reads are completed in 16 or more cycles (2 or more bytes)
6. SPI writes are completed in 16 cycles (2 bytes)
7. R/W first byte contains address & first bit R/W operation / remaining bytes are data
8. Supports single or burst read & writes

RP2350 Example R/W SPI:

```
static inline void cs_low(void) { gpio_put(PIN_CS, 0); }

static inline void cs_high(void) { gpio_put(PIN_CS, 1); }

static void reg_write(uint8_t reg, uint8_t val) {

    uint8_t tx[2] = { (uint8_t)(reg & 0x7F), val };cs_low();
    spi_write_blocking(IMU_SPI, tx, 2);
    cs_high();
}

static void reg_read(uint8_t reg, uint8_t *dst, size_t n) {
    uint8_t hdr = (uint8_t)(reg | 0x80);
    cs_low();
    spi_write_blocking(IMU_SPI, &hdr, 1);
    spi_read_blocking(IMU_SPI, 0x00, dst, n);
    cs_high();
}
```

Writing:

First operation is turning the first byte into a unsigned byte and using a AND operand, it forces the first bit in the reg byte low then the new array tx becomes the full 16 bit write for the ICM. This operation doesn't variable sizes since each write size is constant and specified in the data-sheet.

Reading:

Each read is followed by a single byte write to the IMC, first written byte has its first bit froced high by using a OR operator with 0x80 and the register value, this new value is written to the ICM, the next read operations are the data from the slave device which are in variable sizes. Since each read has different sizes of data the 3rd variable of the function is size.

FIFO Configuration

The IMC-42688-P contains a 2K byte FIFO register accessible via the serial interface. The FIFO configuration register is what determines which data is written into FIFO.

The package structure:

Packet 1	Packet 2	Packet 3	Packet 4
Header (1 byte) ACC data (6 bytes) Temp (1 byte)	header (1 byte) Gyro (6 bytes) Temp (1 byte)	header (1 byte) ACC (6 bytes) Gyro (6 bytes) Temp (1 byte) Time (2 bytes)	Header (1 byte) ACC (6 bytes) Gyro (6 Bytes) Temp (2 bytes) Time (2 bytes) ext (3 bytes)

Package structures are used to transmit different modes, packet 1 transmits as only accelerometer mode, packet 2 is gyro, packet 3 is ACC + gyro + time, packet 3 is same as 23 but with 3 extra bytes. In this case packet 3 will be used since the benefits of packet 4 are negligible in most use cases not involving a well filtered external environment.

Packet 3 content:

Byte	Content
0x00	FIFO Header
0x01	Accel X [15:8]
0x02	Accel X [7:0]
0x03	Accel Y [15:8]
0x04	Accel Y [7:0]
0x05	Accel Z [15:8]
0x06	Accel Z [7:0]
0x07	Gyro X [15:8]
0x08	Gyro X [7:0]
0x09	Gyro Y [15:8]
0x0A	Gyro Y [7:0]
0x0B	Gyro Z [15:8]
0x0C	Gyro Z [7:0]
0x0D	Temperature[7:0]
0x0E	TimeStamp[15:8]
0x0F	TimeStamp[7:0]

FIFO CONFIG1 Registers & setting: (Reg 0x5F : Write 0x47)

BIT	NAME	FUNCTION
4	FIFO_HIRES_EN	0: Default setting; Sensor data have regular resolution 1: Sensor data in FIFO will have extended resolution enabling the 20 Bytes packet with priority on other setting below
3	FIFO_TMST_FSYNC_EN	0: FIFO will only contain ODR timestamp information 1: FIFO can also contain FSYNC time and FSYNC flag for one ODR after an FSYNC event
1	FIFO_GYRO_EN	0: Default setting; Gyroscope data not placed into FIFO 1: Enables gyroscope data packets of 6-bytes to be placed in FIFO
0	FIFO_ACCEL_EN	0: Default setting; Accelerometer data not placed into FIFO 1: Enables accelerometer data packets of 6-bytes to be placed in FIFO

FIFO_HIRES_EN	FIFO_ACCEL_EN	FIFO_GYRO_EN	FIFO_TMST_FSYNC_EN	Header	Packet size
1	X	X	0	8'b_0111_10xx	20 Bytes
1	X	X	1	8'b_0111_11xx	20 Bytes
0	1	1	0	8'b_0110_10xx	16 Bytes
0	1	1	1	8'b_0110_11xx	16 Bytes
0	1	0	X	8'b_0100_00xx	8 Bytes
0	0	1	X	8'b_0010_00xx	8 Bytes
0	0	0 X		No FIFO writes	No FIFO writes

Config (INIT)

The init config is Important to setup the stream, this chapter will contain all register used to setup the INIT of the Gyro. This part consist of setting up how the state machine inside the ICM sensor is working. All registers are listed below, all write reg and values can be found in the final table.

DEVICE_CONFIG (Address: 17 (11h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
4	SPI_MODE	SPI mode selection 0: Mode 0 and Mode 3 (default) 1: Mode 1 and Mode 2
0	SOFT_RESET_CONFIG	Software reset configuration 0: Normal (default) 1: Enable reset After writing 1 to this bitfield, wait 1 ms for soft reset to be effective, before attempting any other register access

REG_BANK_SEL (Address: 118 (76h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: ALL)

BIT	NAME	FUNCTION
2:0	BANK_SEL	Register bank selection 000: Bank 0 (default) 001: Bank 1 010: Bank 2 011: Bank 3 100: Bank 4 101: Reserved 110: Reserved 111: Reserved

FIFO_CONFIG (Address: 22 (16h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:6	FIFO_MODE	00: Bypass Mode (default) 01: Stream-to-FIFO Mode 10: STOP-on-FULL Mode 11: STOP-on-FULL Mode

FIFO_CONFIG1 (Address: 95 (5Fh) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
6	FIFO_RESUME_PARTIAL_RD	0: Partial FIFO read disabled, requires re-reading of the entire FIFO 1: FIFO read can be partial, and resume from last read point
5	FIFO_WM_GT_TH	Trigger FIFO watermark interrupt on every ODR (DMA write) if FIFO_COUNT ≥ FIFO_WM_TH
4	FIFO_HIRES_EN	Enable 3 bytes of extended 20-bits accel, gyro data + 1 byte of extended 16-bit temperature sensor data to be placed into the FIFO
3	FIFO_TMST_FSYNC_EN	Must be set to 1 for all FIFO use cases when FSYNC is used.
2	FIFO_TEMP_EN	Enable temperature sensor packets to go to FIFO
1	FIFO_GYRO_EN	Enable gyroscope packets to go to FIFO
0	FIFO_ACCEL_EN	Enable accelerometer packets to go to FIFO

FIFO_CONFIG2 (Address: 96 (60h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:0	FIFO_WM[7:0]	Lower bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

FIFO_CONFIG3 (Address: 97 (61h) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
3:0	FIFO_WM[11:8]	Upper bits of FIFO watermark. Generate interrupt when the FIFO reaches or exceeds FIFO_WM size in bytes or records according to FIFO_COUNT_REC setting. Interrupt only fires once. This register should be set to non-zero value, before choosing this interrupt source.

INT_CONFIG1 (Address: 100 (64h) | Serial IF: R/W | Reset value: 0x10 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7	—	Reserved
6	INT_TPULSE_DURATION	Interrupt pulse duration 0: Interrupt pulse duration is 100 μ s. Use only if ODR < 4 kHz.(Default) 1: Interrupt pulse duration is 8 μ s. Required if ODR $\geq$ 4 kHz, optional for ODR < 4 kHz.
5	INT_TDEASSERT_DISABLE	Interrupt de-assertion duration 0: The interrupt de-assertion duration is set to a minimum of 100 μ s. Use only if ODR < 4 kHz. (Default) 1: Disables de-assert duration. Required if ODR $\geq$ 4 kHz, optional for ODR < 4 kHz.
4	INT_ASYNC_RESET	User should change setting to 0 from default setting of 1, for proper INT1 and INT2 pin operation

INT_SOURCE0 (Address: 101 (65h) | Serial IF: R/W | Reset value: 0x10 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
6	UI_FSYNC_INT1_EN	0: UI FSYNC interrupt not routed to INT1 1: UI FSYNC interrupt routed to INT1
5	PLL_RDY_INT1_EN	0: PLL ready interrupt not routed to INT1 1: PLL ready interrupt routed to INT1
4	RESET_DONE_INT1_EN	0: Reset done interrupt not routed to INT1 1: Reset done interrupt routed to INT1
3	UI_DRDY_INT1_EN	0: UI data ready interrupt not routed to INT1 1: UI data ready interrupt routed to INT1
2	FIFO_THS_INT1_EN	0: FIFO threshold interrupt not routed to INT1 1: FIFO threshold interrupt routed to INT1
1	FIFO_FULL_INT1_EN	0: FIFO full interrupt not routed to INT1 1: FIFO full interrupt routed to INT1
0	UI_AGC_RDY_INT1_EN	0: UI AGC ready interrupt not routed to INT1 1: UI AGC ready interrupt routed to INT1

GYRO_CONFIG0 | Address: 79 (4Fh) | Serial IF: R/W | Reset value: 0x06 | Clock Domain: SCLK_UI

BITS	NAME	FUNCTION
7:5	GYRO_FS_SEL	Full-scale select for gyroscope UI output: 000: ±2000 dps default 001: ±1000 dps 010: ±500 dps 011: ±250 dps 100: ±125 dps 101: ±62.5 dps 110: ±31.25 dps 111: ±15.625 dps
3:0	GYRO_ODR	Gyroscope ODR: 0000: Reserved 0001: 32 kHz 0010: 16 kHz 0011: 8 kHz 0100: 4 kHz 0101: 2 kHz 0110: 1 kHz default 0111: 200 Hz 1000: 100 Hz 1001: 50 Hz 1010: 25 Hz 1011: 12.5 Hz 1100: Reserved 1101: Reserved 1110: Reserved 1111: 500 Hz

ACCEL_CONFIG0 | Address: 80 (50h) | Serial IF: R/W | Reset value: 0x06 | Clock Domain: SCLK_UI

BITS	NAME	FUNCTION
7:5	ACCEL_FS_SEL	Full-scale select for accelerometer UI output: 000: ±16 g default 001: ±8 g 010: ±4 g 011: ±2 g 100: Reserved 101: Reserved 110: Reserved 111: Reserved
3:0	ACCEL_ODR	Accelerometer ODR: 0000: Reserved 0001: 32 kHz LN 0010: 16 kHz LN 0011: 8 kHz LN 0100: 4 kHz LN 0101: 2 kHz LN 0110: 1 kHz LN default 0111: 200 Hz LP/LN 1000: 100 Hz LP/LN 1001: 50 Hz LP/LN 1010: 25 Hz LP/LN 1011: 12.5 Hz LP/LN 1100: 6.25 Hz LP 1101: 3.125 Hz LP 1110: 1.5625 Hz LP 1111: 500 Hz LP/LN

GYRO_ACCEL_CONFIG0 (Address: 82 (52h) | Serial IF: R/W | Reset value: 0x11 | Clock Domain: SCLK_UI)

BITS	NAME	FUNCTION
7:4	ACCEL_UI_FILT_BW	LN mode accel LPF bandwidth: 0: BW = ODR/2 1: BW = max(400 Hz, ODR)/4 default 2: BW = max(400 Hz, ODR)/5 3: BW = max(400 Hz, ODR)/8 4: BW = max(400 Hz, ODR)/10 5: BW = max(400 Hz, ODR)/16 6: BW = max(400 Hz, ODR)/20 7: BW = max(400 Hz, ODR)/40 8-13: Reserved 14: Low-latency option, Dec2 runs at max(400 Hz, ODR) 15: Low-latency option, Dec2 runs at max(200 Hz, 8×ODR) LP mode: 0: Reserved 1: 1× AVG filter default 2-5: Reserved 6: 16× AVG filter 7-15: Reserved
3:0	GYRO_UI_FILT_BW	LN mode gyro LPF bandwidth: 0: BW = ODR/2 1: BW = max(400 Hz, ODR)/4 default 2: BW = max(400 Hz, ODR)/5 3: BW = max(400 Hz, ODR)/8 4: BW = max(400 Hz, ODR)/10 5: BW = max(400 Hz, ODR)/16 6: BW = max(400 Hz, ODR)/20 7: BW = max(400 Hz, ODR)/40 8-13: Reserved 14: Low-latency option, Dec2 runs at max(400 Hz, ODR) 15: Low-latency option, Dec2 runs at max(200 Hz, 8×ODR)

PWR_MGMT0 (Address: 78 (4Eh) | Serial IF: R/W | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
5	TEMP_DIS	0: temperature sensor enabled default 1: temperature sensor disabled
4	IDLE	If 1, RC oscillator stays powered even if accel/gyro are off. Normally 0: when accel/gyro are off, chip goes OFF because RC oscillator also powers off.
3:2	GYRO_MODE	00: gyro off default 01: gyro standby 10: Reserved 11: gyro Low Noise mode Note: gyro must stay ON ≥ 45 ms. When leaving OFF, do not write registers for 200 μ s.
1:0	ACCEL_MODE	00: accel off default 01: accel off 10: accel Low Power mode 11: accel Low Noise mode Note: when leaving OFF, do not write registers for 200 μ s.

SIGNAL_PATH_RESET (Address: 75 (4Bh) | Serial IF: W/C | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7	—	Reserved
6	DMP_INIT_EN	When this bit is set to 1, the DMP is enabled
5	DMP_MEM_RESET_EN	When this bit is set to 1, the DMP memory is reset
4	—	Reserved
3	ABORT_AND_RESET	When this bit is set to 1, the signal path is reset by restarting the ODR counter and signal path controls
2	TMST_STROBE	When this bit is set to 1, the time stamp counter is latched into the time stamp register. This is a write on clear bit.
1	FIFO_FLUSH	When set to 1, FIFO will get flushed.
0	—	Reserved

Config Writes:

Values To Write	Config Description
Dev reg Write: DEVICE_CONFIG (0x11 0x01) / 00000001 BANK_SEL (0x76 0x00) / 00000000	Enables the SPI mode & selects bank 1
FIFO reg Write Values: FIFO_CONFIG (0x16 0x40) / 01000000 FIFO_CONFIG_1 (0x5F 0x47) / 01000111 FIFO_CONFIG_2 (0x60 0x20) / 00100000 FIFO_CONFIG_3 (0x61 0x00) / 00000000	FIFO_CONFIG is used to configure the stream, This application will use Stream to fifo Mode. FIFO_CONFIG_1 enables Acc[0], gyro[1], temp[2] and FIFO_RESUME_PARTIAL_RD[1] this registers main use case is to enable sensors IFO_RESUME_PARTIAL_RD is used to resume from last read giving a cleaner stream. FIFO_CONFIG_2 is used as a water mark, the watermark is what generates the interrupt for addressing that the fifo is full increasing decreases the workload on SPI but increases the latency. FIFO_CONFIG_3 is used for for watermarks larger than 256 bytes.
INT Reg Write: INT_CONFIG1 (0x64 0x00) / 00000000 INT_SOURCE0 (0x65 0x04) / 00001000	This setting sets the interrupt FIFO pulse to last 100ms, for 1khz ODR its ideal. The Source0 interrupt is set to Int 1 when FIFO is full.
GYRO ACC REG R/W: GYRO_CONFIG0 (0x4F 0x06) / 00000110 ACCEL_CONFIG0 (0x50 0x06) / 00000110 GYRO_ACCEL_CONFIG0 (0x52 0x11) / 00010001	Gyro set to 1Khz, Accel set to 1Khz, & filtering is set to default. These values may be changed during testing for optimal performance.
PWR SIG Reg R/W: PWR_MGMT0 (0x4E 0x0F) / 00001111 SIGNAL_PATH_RESET (0x4B 0x02) / 00000010	Power management enables the Gyro & accelerometer in Low noise mode & temperature is enabled. Singal path enables FIFO flush which sets a bit to 1 when fifo gets flushed.

FIFO_COUNTH (Address: 46 (2Eh) | Serial IF: R | Reset value: 0x00 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:0	FIFO_COUNT[15:8]	High bits, count indicates the number of records or bytes available in FIFO according to FIFO_COUNT_REC setting. Reading this byte latches the data for both FIFO_COUNTH and FIFO_COUNTL.

INT_STATUS (Address: 45 (2Dh) | Serial IF: R/C | Reset value: 0x10 | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
6	UI_FSYNC_INT	This bit automatically sets to 1 when a UI FSYNC interrupt is generated. The bit clears to 0 after the register has been read.
5	PLL_RDY_INT	This bit automatically sets to 1 when a PLL Ready interrupt is generated. The bit clears to 0 after the register has been read.
4	RESET_DONE_INT	This bit automatically sets to 1 when software reset is complete. The bit clears to 0 after the register has been read.
3	DATA_RDY_INT	This bit automatically sets to 1 when a Data Ready interrupt is generated. The bit clears to 0 after the register has been read.
2	FIFO_THS_INT	This bit automatically sets to 1 when the FIFO buffer reaches the threshold value. The bit clears to 0 after the register has been read.
1	FIFO_FULL_INT	This bit automatically sets to 1 when the FIFO buffer is full. The bit clears to 0 after the register has been read.
0	AGC_RDY_INT	This bit automatically sets to 1 when an AGC Ready interrupt is generated. The bit clears to 0 after the register has been read.

FIFO_DATA (Address: 48 (30h) | Serial IF: R | Reset value: 0xFF | Clock Domain: SCLK_UI)

BIT	NAME	FUNCTION
7:0	STEP_CNT[15:8]	FIFO data port

Reg / Size	Description
Reg Data_size (Bytes) FIFO_COUNTH (0x46 2) FIFO_DATA (0x30 16) INT_STATUS (0x45 1)	FIFO_COUNTH is a indicator byte used to indicate the number of bytes in the FIFO. FIFO_DATA is the actual data in the fifo holding gyro, acc, temp & time. INT_STATUS clears previous flags & stale interrupt, used after init.

Write:

```
reg_write(REG_DEVICE_CONFIG, 0x01)
reg_write(REG_BANK_SEL, 0x00);
reg_write(REG_FIFO_CONFIG, 0x40);
reg_write(REG_GYRO_CONFIG0, 0x06);
reg_write(REG_ACCEL_CONFIG0, 0x06);
reg_write(REG_GYRO_ACCEL_CONFIG0, 0x11);
reg_write(REG_FIFO_CONFIG1, 0x47);
reg_write(REG_FIFO_CONFIG2, (uint8_t)(FIFO_WATERMARK_BYTES & 0xFF));
reg_write(REG_FIFO_CONFIG3, 0x00);
reg_write(REG_INT_CONFIG1, 0x00);
reg_write(REG_INT_SOURCE0, 0x04);
reg_write(REG_PWR_MGMT0, 0x0F);
reg_write(REG_SIGNAL_PATH_RESET, 0x02);
```

Read:

Which ICM-42688-P settings are best for inertial navigation (4 Technical report)

Of the four parts, the ICM-42688-P is the clear pick for a strapdown INS – it has the lowest gyro noise, the tightest offset temp stability, and the 32 kHz ODR with RTC time-stamping. But the headline specs only get you half the way; the part's actual INS performance is set by how you configure it. Here's what matters, in order of impact on navigation error growth:

1. Gyro full-scale – trade measurement range for effective resolution. The "16/19-bits" resolution is a fixed-point format: full-scale stays 2^{19} counts regardless of FSR, so the scale factor per LSB scales linearly with the range you select. Angle error from quantization is roughly $\sigma_q \approx (\text{FSR} / 2^{19}) \cdot (1/\sqrt{12})$ per sample; at ± 250 dps that's ≈ 0.004 mdps/LSB, vs 0.03 mdps/LSB at ± 2000 dps. For platform stabilization or slow-moving navigation, use **± 250 or ± 500 dps** (GYRO_CONFIG0, register 0x4F). Only go to ± 2000 dps if you have genuine body rates above a few hundred dps – high vibration environments or fast attitude maneuvers – because 2.8 mdps/ $\sqrt{\text{Hz}}$ noise density matters more than quantization until FSR gets large.

2. Gyro ODR and anti-alias filtering. The 32 kHz ODR (GYRO_CONFIG0, 32 kHz mode) is valuable for two reasons: it averages the integrated rate over shorter windows, and more importantly it lets you push the **gyro anti-alias filter (AAF, bank 1, registers 0x0C–0x0D)** low enough to reject structural vibration before decimation. For a vehicle or UAV, set AAF corner $\sim$ half your decimated rate. Note that the accel is limited to 32 kHz as well, but the usable signal bandwidth tops out around a few kHz – don't confuse ODR with bandwidth.

3. Offset stability vs. bias calibration. The ± 5 mdps/ $^{\circ}\text{C}$ spec is the table's best by a factor of 2, but understand what it is and isn't: it's the *slope* of zero-rate offset vs. temperature. In an INS, attitude error from an uncorrected gyro bias b grows linearly: $\Delta\theta(t) \approx b \cdot t$ (plus the noise-driven random-walk term, $\Delta\theta \sim N \cdot \sqrt{(2T)}$ with $N = 2.8$ mdps/ $\sqrt{\text{Hz}} \approx 0.162^{\circ}/\sqrt{\text{hr}}$ ARW-class performance). A 5 mdps/ $^{\circ}\text{C}$ slope means a 20°C swing uncalibrated injects 100 mdps $\approx 0.34^{\circ}/\text{s}$ of error – that dominates everything. So: temperature-compensate with an offline calibration (store bias as a function of temperature in firmware), or keep the IMU thermally controlled. With calibration you reduce the effective bias to the hysteresis level, which is what separates good dead reckoning from drift within minutes.

4. Accel range and noise anisotropy. Use **± 16 g** (ACCEL_CONFIG0) – you don't need ± 32 g on this part, and staying at ± 16 g maximizes your effective resolution. Note the spec sheet's quirk: X-axis noise density is $65 \mu\text{g}/\sqrt{\text{Hz}}$ vs 70 for Y and Z. For a Z-up or X-forward navigation frame, the difference is marginal, but it's worth mounting the part so the lowest-noise axis (X) aligns with the most velocity-critical direction. The accel bias stability isn't in this table – check the full datasheet's bias repeatability specs, because accel bias integrates directly into velocity error, $\Delta v \approx b_a \cdot t$, and tilt misalignment couples gravity (~ 2.4 mm/s of pseudo-velocity error per milliradian of tilt, per g).

5. RTC and time alignment. This is the unsung advantage of the -P variants: the 32 kHz sample clock synchronized to the internal RTC (via the TMST registers) gives you sub-microsecond-consistent time stamps on every FIFO sample. For INS/EKF fusion with GNSS, odometry, or vision, timestamp error Δt on a rotating body creates apparent attitude error $\omega \cdot \Delta t$ – at $500^{\circ}/\text{s}$, 1 ms of time error is already 0.5° . Read TMST_S0 (0x62) and TMST_S1 (0x63) in user bank 0 to get the RTC-synced FIFO timestamps.

Practical register setup for navigation (bank 0):

- PWR_MGMT0: accel + gyro on in LN (low-noise) mode – avoid LP (low-power) mode; it uses duty-cycled sensing with worse bias

- GYRO_CONFIG0: ± 250 –500 dps, 32 kHz (or 8 kHz if power budget forces it; 32 kHz if you want the AAF margin)

- ACCEL_CONFIG0: ± 16 g, 32 kHz

- GYRO_CONFIG1: UI filter order 3rd (lowest group delay for real-time fusion)

- Bank 1, GYRO_X/Y/Z_AAF + GYRO_CONFIG_STATIC6–9: AAF corners matched to your signal band

- FIFO_CONFIG / TMST_S0–S1: enable RTC time stamps and stream 20-byte packets (accel, gyro, temp, time)

One caveat from the datasheet itself: you can only change ODR, FSR, and operating mode while the sensors are running – every other register write requires both accel and gyro off first (register 12.9), so build your init sequence to set all filters and configs before enabling sensors, or accept a re-init cycle when changing them.

The one thing no register setting fixes: the gyro's long-term bias stability (the "instability" term in an Allan deviation) isn't specified in this table, so if you're doing minutes-long dead reckoning without GNSS aiding, plan on periodic zero-velocity or magnetometer updates regardless of how well you calibrate temperature.